

Wormholes and the Possibility of Time Travel

Emily Tieu

March 30, 2026

Defining Wormholes

Wormholes: Multiply-connected spatial geometry with non-trivial topologies that connect two distant regions.

- Hypothetical solutions to the Einstein field equations

$$G^{\mu\nu} = 8\pi T^{\mu\nu},$$

where $G^{\mu\nu} = R^{\mu\nu} - \frac{1}{2}Rg^{\mu\nu}$ is the Einstein tensor and $T^{\mu\nu}$ is the stress-energy tensor

- While the distance between mouths may be infinite, the distance through the wormholes may be arbitrarily small

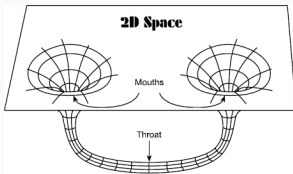


Figure 1: Simple wormhole

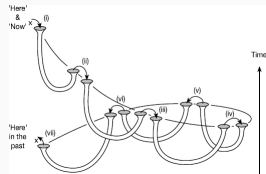


Figure 2: Multiple adjoining wormholes

Evolution of Wormhole Models

Schwarzschild Solution

Beginning with **Schwarzschild's solution**:

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi)^2,$$

the region containing the Schwarzschild radius $r = 2M$ was realized to be a coordinate singularity, known as the event horizon.

- Any object within this horizon would be required to travel faster than the speed of light to escape

Einstein–Rosen Bridge

The coordinate transformation with $u^2 = r - 2M$ was performed to derive the **Einstein–Rosen bridge**:

$$ds^2 = -4(u^2 + 2M)du^2 - (u^2 + 2M)^2 d\Omega^2 + \frac{u^2}{u^2 + 2M} dt^2,$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$.

Gravitational Collapse: However, there is no repulsive force that keeps the throat open. Since the metric term $(1 - \frac{2M}{r})^{-1}$ blows up near the event horizon, the throat radius collapses.

$\Rightarrow$ The Einstein–Rosen Bridge is **non-traversable**.

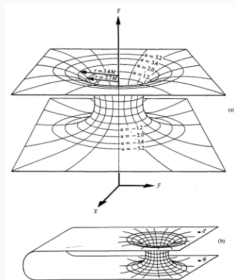


Figure 3: ER bridge

Morris–Thorne Model

In the Schwarzschild coordinates (t, r, θ, ϕ) , the geometry of a static, spherically symmetric wormhole can be represented by the

Morris–Thorne metric

$$ds^2 = -e^{2\varphi(r)} dt^2 + \frac{dr^2}{1 - \frac{b(r)}{r}} + r^2(d\theta^2 + \sin^2 \theta d\phi^2),$$

where:

- $\varphi(r)$ is the gravitational redshift function, and must be finite s.t.
 - $g_{tt} = -e^{2\varphi(r)}$ never vanishes
 - an event horizon does not form
 - signals and observers can pass freely without encountering infinite gravitational redshift
- $b(r)$ determines the spatial geometry of the wormhole
- r is the radial coordinate, and must range from r_0 , the wormhole's throat, to $+\infty$

Morris–Thorne Model

Criteria for traversability: The shape function $b(r)$ must furthermore satisfy the following conditions:

- *Throat condition:* $b(r_0) = r_0$, allowing $1 - \frac{b(r)}{r} > 0$ for $r_0 > r$
- *Flaring-out condition:* $b'(r_0) < 1$
- *Asymptotic flatness condition:* As $r \rightarrow \infty$, $\frac{b(r)}{r} \rightarrow 0$

Null Energy Condition: $T_{\mu\nu}k^\mu k^\nu \geq 0$ for every null vector field k^μ .

However, the flaring-out condition imposes a direct **violation**:

$$\begin{aligned}T_{\mu\nu}k^\mu k^\nu &\propto \frac{b'(r) - b(r)/r}{r^2} \\ \implies T_{\mu\nu}k^\mu k^\nu &\propto \frac{b'(r_0) - 1}{r_0^2} && \text{(at the throat)} \\ \implies T_{\mu\nu}k^\mu k^\nu &< 0\end{aligned}$$

$\therefore$ Some form of exotic matter with negative energy density must exist to maintain flaring geometry.

Implications

Closed Timelike Curves

For a particle following a geodesic parameterized by proper time τ along trajectory $\gamma(\tau)$,

$$g_{\mu\nu} \frac{d\gamma^\mu}{d\tau} \frac{d\gamma^\nu}{d\tau} < 0, \forall \tau \in [\tau_1, \tau_2],$$

where $\tau_1 < \tau_2$ and $\gamma(\tau_1) = \gamma(\tau_2)$ defines a closed timelike curve.

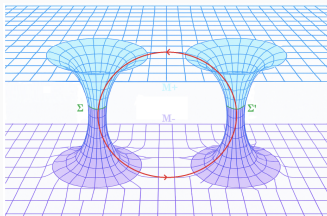


Figure 4: Geometry permitting a CTC, shown in red

The addition of a second one-way (Einstein–Rosen) wormhole would complete a CTC.

Closed Timelike Curves

For two spacetime regions $\mathcal{M}_+$ and $\mathcal{M}_-$, define two Einstein–Rosen bridges made traversable in Eddington–Finkelstein coordinates:

- Incoming metric $g^{(+)}$:

$$ds_+^2 = \left(1 - \frac{\alpha}{r}\right) dt^2 - \left(1 + \frac{\alpha}{r}\right) dr^2 - \frac{2\alpha}{r} dr dt - r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

- Outgoing metric $g^{(-)}$:

$$ds_-^2 = \left(1 - \frac{\alpha}{r}\right) dt^2 - \left(1 + \frac{\alpha}{r}\right) dr^2 - \frac{2\alpha}{r} dr dt - r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

Then the derivative of coordinate time t w.r.t. proper time τ appears to change sign:

- Under $g^{(+)}$: $\frac{dt}{d\tau} > 0$ (evolution towards the future)
- Under $g^{(-)}$: $\frac{dt}{d\tau} < 0$ (evolution towards the past)

Thus, there must be a time inversion at the throat, i.e., $\frac{dt}{d\tau}|_{t_0} = 0$.

Time Travel?

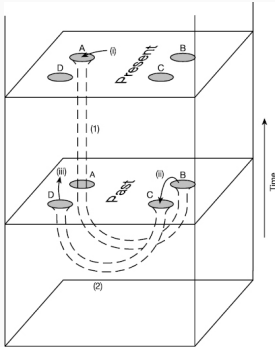


Figure 5: A Roman time machine using two wormholes

- CTCs enable a particle to return to its original spacetime point, essentially revisiting its own past
- Time dilation + wormhole geometry allows global loop back in spacetime

Hawking's chronology protection conjecture: The laws of physics prevent closed timelike curves (time travel) on macroscopic scales.

References

- [1] J. Al-Khalili, *Black holes, Wormholes, and Time Machines*, 2nd ed. Boca Raton, FL: Taylor and Francis, 2012.
- [2] P. H. R. S. Moraes, R. A. C. Correa, and R. V. Lobato, "Analytical General Solutions for Static Wormholes in $f(R, T)$ Gravity," *Journal of Cosmology and Astroparticle Physics*, vol. 2017, no. 7, pp. 29–29, July 2017.
- [3] V. Venkatesha, N. S. Kavya, and P. K. Sahoo, "Geometric structures of Morris-Thorne wormhole metric in $f(L, m)$ gravity and energy conditions," *Physical scripta*, vol. 98, no. 6, p. 65020, June 2023.
- [4] H. Zejli, "Closed timelike curves in η -symmetric wormholes," *International journal of modern physics. D, Gravitation, astrophysics, cosmology*, vol. 34, no. 11, Aug. 2025.